

TELECOM CUSTOMER CHURN ANALYSIS REPORT

Identifying Churn Drivers & Revenue Impact

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Dataset: Telecom Customer Churn Dataset
Tool: MySQL Workbench
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Executive Summary

This report presents a comprehensive SQL-driven analysis of customer churn for a telecom company. Using a dataset of 2,666 customers, the analysis identifies the key behavioural, plan-based, and usage-related drivers of churn, quantifies revenue impact, and highlights high-risk customer segments to guide targeted retention strategies.

2,666 Total Customers Full dataset	388 Churned Customers 14.55% churn rate	2,278 Retained Customers 85.45% retained	15 High-Risk Customers Needs urgent action
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Key Findings at a Glance

- International plan subscribers churn at nearly 4x the rate of non-subscribers (43.70% vs 11.27%).
- Customers making 5+ service calls churn at 61% — a critical distress signal.
- High day-charge users churn at 38.15%, suggesting pricing dissatisfaction.
- Texas leads revenue loss with \$1,089 lost to churn, across 10 high-loss states.
- Voicemail plan subscribers show significantly lower churn (8.87% vs 16.71%).

1. Dataset Overview

The dataset contains 2,666 telecom customer records with 20 attributes spanning subscription plans, usage patterns, charges, and churn status. The table below summarises the key fields used in this analysis.

Column	Description	Type
state	US state of the customer	Text
account_length	Duration of account in days	Numeric
international_plan	Subscribed to international plan?	Yes / No
voice_mail_plan	Subscribed to voicemail plan?	Yes / No
customer_service_calls	Number of calls made to support	Numeric
total_day_charge	Total daytime call charges (\$)	Numeric
total_intl_charge	Total international call charges (\$)	Numeric
churn	Did the customer leave? (Target)	True / False

2. Overall Churn Overview

2.1 Total Customers & Churn Distribution

Churn Status	Customer Count	Percentage
Not Churned (False)	2,278	85.45%
Churned (True)	388	14.55%
Total	2,666	100%



Churn Rate Context

- The industry average churn rate for telecom companies is typically 1.5%–3% monthly (or ~20-25% annually). At 14.55%, this company's churn is a significant concern requiring immediate strategic action.
- For every 100 customers, approximately 15 are leaving — highlighting a retention gap that, if closed even partially, could drive meaningful revenue recovery.

2.2 Average Charges — Churned vs. Retained

Churn Status	Avg Day Charge	Avg Evening Charge	Avg Night Charge	Avg Intl Charge
Retained	\$29.77	\$16.90	\$9.02	\$2.74
Churned	\$34.88	\$17.80	\$9.24	\$2.92



Charge Pattern Insight

- Churned customers consistently pay more across all time periods, with day charges being the most significant differentiator (+\$5.11 on average).
- This suggests a price-sensitivity problem: high spenders feel they are not receiving value proportional to cost and are switching to competitors.

2.3 Average Usage Minutes — Churned vs. Retained

Churn Status	Avg Day Mins	Avg Eve Mins	Avg Night Mins	Avg Intl Mins
Retained	175.10	198.85	200.46	10.14
Churned	205.18	209.39	205.31	10.82



Usage Insight

- Churned customers are heavier users across all segments — they use more minutes yet still leave. This is a classic 'heavy user churn' pattern, often caused by perceived billing unfairness.
- Despite consuming significantly more minutes, these customers are not receiving loyalty rewards or personalised pricing, making them prime targets for competitors offering better value plans.

3. Churn by Subscription Plan

3.1 International Plan

International Plan	Total Customers	Churned	Churn Rate
Yes	270	118	43.70%
No	2,396	270	11.27%

 Critical Finding — International Plan

- International plan subscribers churn at 43.70% — nearly 4x the rate of non-subscribers (11.27%). This is the single strongest churn predictor in the dataset.
- Root cause hypothesis: International call rates may be uncompetitive. Customers sign up expecting value but quickly switch after experiencing high bills relative to competitors.
- Recommendation: Audit international plan pricing, introduce tiered international packages, and proactively reach out to international plan subscribers before they churn.

3.2 Voice Mail Plan

Voice Mail Plan	Total Customers	Churned	Churn Rate
No	1,933	323	16.71%
Yes	733	65	8.87%

 Voicemail Plan as a Retention Tool

- Customers with a voicemail plan churn at nearly half the rate (8.87%) of those without one (16.71%).
- Voicemail subscribers likely have greater product engagement and feature utilization, which increases switching costs and loyalty.
- Recommendation: Promote voicemail plan adoption — consider offering it as a free trial or bundled feature, especially for at-risk customer segments.

4. Churn by Customer Service Calls

4.1 Churn Rate by Number of Calls

Service Calls	Total Customers	Churned	Churn Rate
0	555	79	14.23%
1	945	99	10.48%
2	608	62	10.20%
3	348	37	10.63%
4	133	64	48.12%
5	49	29	59.18%
6	17	10	58.82%
7	8	5	62.50%
8	1	1	100.00%
9	2	2	100.00%

4.2 Churn Rate by Call Group

Call Group	Total Customers	Churned	Churn Rate
1–2 Calls	1,553	161	10.37%
0 Calls	555	79	14.23%
3–4 Calls	481	101	21.00%
5+ Calls	77	47	61.04%

The '4-Call Threshold' — A Critical Alert Signal

- Churn remains relatively stable at around 10-14% for customers making 0-3 calls. However, at 4 calls, churn jumps sharply to 48% — nearly 5x higher.
- Customers making 5+ calls churn at over 61%, indicating unresolved service issues are a leading cause of departure.
- Recommendation: Implement an automated alert in the CRM system when a customer reaches their 3rd service call. Trigger proactive outreach with a retention specialist before they reach 4 calls.

5. Churn by Account Length & Usage

5.1 Churn by Account Length Group

Account Length Group	Total Customers	Churned	Churn Rate
0–50 Days	278	37	13.31%
51–100 Days	1,063	151	14.21%
101–150 Days	1,051	155	14.75%
150+ Days	274	45	16.42%

💡 Churn Grows with Tenure — A Loyalty Paradox

- Churn rate increases slightly as account length grows, with the highest churn among the longest-tenured customers (150+ days at 16.42%).
- This counter-intuitive pattern suggests that the company is NOT rewarding loyal customers. Long-standing customers may feel undervalued, especially if they are paying more for the same plans as new customers.
- Recommendation: Introduce a loyalty programme that offers incremental benefits at tenure milestones (e.g., 6 months, 1 year). Loyal customers should be the easiest to retain.

5.2 Churn by Total Day Charge Group

Day Charge Group	Total Customers	Churned	Churn Rate
Medium Charge	1,527	147	9.63%
Low Charge	738	88	11.92%
High Charge	401	153	38.15%

💡 High Billing = High Churn Risk

- High-charge customers churn at 38.15% — nearly 4x the rate of medium-charge customers. These are the company's most valuable customers by revenue, yet they are the most likely to leave.
- Recommendation: Identify high-charge customers proactively and offer them personalised loyalty discounts, data bundles, or plan upgrades before they decide to leave.

5.3 Churn by International Usage Group

International Usage	Total Customers	Churned	Churn Rate
No International Usage	15	0	0.00%
Low Usage	1,240	156	12.58%
Medium Usage	1,315	213	16.20%
High Usage	96	19	19.79%

International Usage Drives Churn Incrementally

- As international usage rises, so does churn. Medium and high international users have meaningfully higher churn rates than low users.
- Combined with the international plan finding, this suggests that customers who use international services frequently are underserved by the current plan structure.

6. Geographic Churn Analysis

6.1 Top 10 States by Churn Rate

State	Total Customers	Churned	Churn Rate
TX	55	16	29.09%
NJ	50	14	28.00%
AR	47	11	23.40%
MD	60	14	23.33%
MS	48	11	22.92%
SC	49	11	22.45%
ME	49	11	22.45%
MI	58	13	22.41%
PA	36	8	22.22%
NV	61	13	21.31%

Geographic Churn Clusters

- Texas leads with a 29.09% churn rate — double the national average — making it the highest-priority state for targeted retention campaigns.
- New Jersey (28%) and Maryland (23.33%) are significant churn hotspots in the Northeast corridor.
- 9 of the top 10 churning states exceed 20% churn rate, suggesting possible regional competitive pressure from local carriers or infrastructure/coverage issues.
- Recommendation: Investigate common patterns in these states — are there specific competitors, coverage gaps, or plan pricing issues driving higher dissatisfaction?

7. Revenue Impact Analysis

7.1 Estimated Revenue Lost to Churn by State (Top 10)

Revenue lost is estimated based on total charges (day + international) of churned customers per state.

State	Churned Customers	Est. Revenue Lost (\$)
TX	16	\$1,089.43
MD	14	\$969.73
NJ	14	\$966.31
NV	13	\$861.66
MN	13	\$857.66
MI	13	\$829.81
CT	11	\$770.31
OH	10	\$710.71
SC	11	\$707.65
NY	12	\$702.71

 Revenue Loss Insight

- Texas alone accounts for over \$1,000 in estimated lost revenue, making it both the highest-churn and highest revenue-loss state.
- The top 10 states collectively represent significant recoverable revenue. Even retaining 30-40% of these churned customers would deliver substantial financial impact.
- Recommendation: Prioritise retention spend in TX, MD, NJ, and NV as these states offer the highest return on retention investment.

8. High-Risk Customer Segments

8.1 Critical High-Risk Segment (Combined Risk Factors)

Criteria: customer service calls of more than 4 having international plan and ended their relationship with a company within a given period.

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Customers Meeting ALL Criteria
Already churned — use for model training

What This Tells Us

- These 15 customers represent the 'perfect storm' of churn risk: they have an expensive international plan, had multiple unresolved support issues, and ultimately left.
- This profile should form the core of a predictive churn model — customers who share 2 of 3 of these characteristics should be flagged immediately for proactive retention.

8.2 Broader At-Risk Customer List

Criteria: customer service calls ≥ 4 OR total day charge > 40 OR total international charge > 3

Sample of at-risk customers identified (top 10 shown):

State	Acct Length	Intl Plan	VM Plan	Day Charge	Intl Charge	Svc Calls	Churn
MN	152	Yes	Yes	\$40.38	\$3.83	9	True
NJ	53	No	No	\$38.86	\$2.73	9	True
NH	103	No	No	\$25.60	\$2.81	8	True
KY	122	No	Yes	\$43.13	\$2.48	7	False
HI	149	No	No	\$28.32	\$2.21	7	False
OH	93	No	Yes	\$25.89	\$3.05	7	False
MI	146	No	No	\$25.38	\$2.13	7	True
AR	115	No	No	\$23.68	\$2.51	7	True
HI	99	No	No	\$23.07	\$2.38	7	True
AL	86	No	No	\$21.81	\$3.29	7	True

Actionable Observations from At-Risk List

- Several customers with 7-9 service calls have NOT yet churned (e.g., KY, HI, OH) — these are the most urgent retention targets as they show all the warning signs.

- High day charges (>\$40) combined with multiple service calls indicates customers who are both high-value AND highly frustrated — a combination that demands immediate attention.
- Recommendation: Export this list monthly and assign each customer to a retention specialist for personalised outreach.

9. Strategic Recommendations

Priority	Action	Target Segment	Expected Impact
 Critical	Redesign international plan pricing	Int'l plan subscribers	Reduce 43.7% churn rate
 Critical	CRM alert at 3rd service call	High service call customers	Prevent 5+ call escalation
 High	Proactive outreach to high-charge users	Top day charge customers	Retain high-value base
 High	State-specific retention campaigns	TX, NJ, MD, NV	Recover lost revenue
 Medium	Promote voicemail plan adoption	Non-voicemail subscribers	Lower overall churn rate
 Medium	Launch loyalty milestones programme	100+ day account holders	Address tenure paradox
 Long-term	Build predictive churn ML model	All active customers	Flag risk before churn

10. Conclusion

This SQL-based churn analysis has uncovered clear, actionable patterns within the telecom customer dataset. The overall churn rate of 14.55% is driven by a concentrated set of risk factors that, once addressed, can significantly improve customer retention.

The most critical findings are:

- The international plan is the strongest churn predictor (43.70% churn rate) and requires urgent pricing review.
- Customer service call volume is a direct proxy for dissatisfaction — every additional call beyond 3 dramatically increases churn likelihood.
- High-spend customers are paradoxically the most likely to churn, signalling a perceived value gap that loyalty offers can bridge.
- Geographic and voicemail plan patterns provide clear segmentation opportunities for targeted retention campaigns.

By implementing the recommendations in this report — particularly proactive CRM interventions, international plan restructuring, and loyalty rewards for long-tenure customers — the company can realistically reduce churn by 30–40%, translating directly into recovered revenue and a stronger customer lifetime value.

Data tells a story. This analysis finds the plot twists before customers write the ending.
